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HISTORY · Changes in administration, economy, society and culture under the British

UNDERSTAND IT FIRST

What is it? After the **Revolt of 1857** the British Crown took India from the Company (**1858**) and ruled it for **89 more years**. This topic asks **what changed** under that rule: in **how India was governed**, in **how its economy worked**, and in **how society and culture were shaped**. The key idea: the changes were made **for British benefit first**; some (railways, English education, modern law) also had **unintended, lasting effects on India**. Learn it as **Administration → Economy → Education and press → Society and culture → Andhra link → Impact**.

How India was governed after 1857

1. Changes in administration

Measure	Details
Government of India Act, 1858	**Crown rule** replaced the Company; the **Board of Control and Court of Directors were abolished**; the **Governor-General became Viceroy** (**Lord Canning**, first Viceroy); a **Secretary of State for India** (a British Cabinet minister) was helped by a **Council of 15**; India was ruled from London.
Queen's Proclamation, 1 November 1858	Read at **Allahabad** by Canning: **no more annexation** of Indian states, **religious tolerance**, **equal and impartial service** for all subjects, respect for **treaties and customs**; it was the Crown's **promise**, often broken. (Sometimes called the **'Magna Carta of Indian liberties'**.)
Indian Councils Acts	**1861**: Viceroy's Council became a body with **portfolios** (Canning's **portfolio system**); the **Legislative Council** got **6–12 additional members** (nominated, a few **Indians** like the Raja of Benares), powers **restored to Madras and Bombay**; Viceroy could issue **ordinances**. **1892**: more additional members, power to **discuss the budget** and ask questions, **indirect election** in principle.
Army	After 1857 (**Peel Commission, 1859**): **more European troops** (about 1 to 2 in the Bengal army, 2 to 5 in Madras and Bombay), **artillery in European hands only**, recruits from the **'martial races'** (Sikhs, Gurkhas, Punjabi Muslims) and soldiers mixed by caste and region to prevent unity.
Princely States	Policy of **no annexation**; rulers became **loyal 'props'** of the Empire (about **560** states, one-third of the area) under **paramountcy** and a British **Resident**.
Local government and finance	**Lord Mayo (1870)**: **financial decentralisation** gave provinces fixed grants for some services. **Lord Ripon (1880–84)**: **Resolution of 1882** on **local self-government** (elected local boards), hence the **'father of local self-government'**; also repealed the **Vernacular Press Act (1882)**, passed the **first Factory Act (1881)**, set up the **Hunter Education Commission (1882)**, and

	faced the **Ilbert Bill (1883)** controversy (Indian judges trying Europeans). **Lord Lytton (1876–80)** passed the **Vernacular Press Act and Arms Act (1878)**.

2. Changes in the economy

The colonial economy

UNDERSTAND IT FIRST

Why this matters: before the British, India was a **leading exporter of cloth and spices**. **Under** colonial rule it became a **supplier of raw materials** (cotton, jute, indigo, tea) and a **market for British factory goods**. This is called a **colonial economy**: the **economy served the ruler's interest**.

Feature	Details
Drain of wealth	A part of India's wealth went to Britain **with no return**: **'Home Charges'** (interest on debt, pensions, salaries, costs of the India Office), profits and savings of Europeans. **Dadabhai Naoroji** first stated it in **'England's Debt to India' (1867)** and in **'Poverty and Un-British Rule in India' (1901)**; **R.C. Dutt** in **'Economic History of India' (1901–03)**; also **M.G. Ranade** and **G. Subramania Iyer**.
De-industrialisation	Cheap **machine-made cloth** from Britain with **low duties** and **high duties on Indian goods** ruined **weavers and artisans** (e.g. Dhaka muslin); **fewer jobs** pushed people **back onto the land**; **ruralisation**.
Commercialisation of agriculture	Farmers grew **cash crops** (**indigo, cotton, jute, opium, tea, sugarcane, tobacco**) for **export** because **taxes had to be paid in money**; **less land for food crops**; **moneylender debt** grew; **indigo growers** in Bengal rose in the **Indigo Revolt (1859–60)** (Dinabandhu Mitra's play **'Nil Darpan'**).
Famines	Heavy land revenue, export of grain and poor relief made famines deadly: **Great Famine 1876–78** (millions died; worst in **Madras Presidency, Mysore, Hyderabad and the Bombay Deccan**; **Lytton** was Viceroy); **Famine Commission 1880** under **Richard Strachey** led to the **Famine Code (1883)**; later **Lyall (1898)** and **MacDonnell (1901)** commissions; famines of **1896–97, 1899–1900**.
Railways, post, telegraph	First line **Bombay–Thane (1853)**, then **Calcutta–Raniganj (1854)** and **Madras (Royapuram–Wallajah Road, 1856)**. Built by **British companies with a guaranteed 5% return** paid from Indian revenue; most **steel, engines and coal came from Britain**; routes linked **ports, raw-material areas and army posts**. **Good effects:** a **national market**, movement of people and ideas, relief of famine areas. **Bad effects:** it **did not build Indian industry**.
Modern industry	Mostly **Indian-owned, small** at first: **cotton mill, Bombay (1854, Cowasji Davar)**, **jute mill, Rishra (1855)**, coal at **Raniganj**, later **Tata Iron and Steel, Jamshedpur (1907)**; no **heavy industry** and a **'free trade'** policy that gave no protection.

3. Education and the press

Step	Details
Before 1835	**Orientalists** (Warren Hastings, **William Jones**: Calcutta Madrasa 1781, Asiatic Society 1784, Sanskrit College Benares 1791) backed **Indian classical learning**; **Charter Act 1813**: ₹1 lakh; the **Orientalist–Anglicist controversy**.
Macaulay's Minute, 1835	**Anglicists** won: **English as medium** for higher education to create a class **'Indian in blood and colour, English in taste, opinions, morals and intellect'**; **Bentinck's Resolution (March 1835)**; the **'downward filtration theory'** (educate the few, and learning would filter down).
Wood's Despatch, 1854	**'Magna Carta of English education in India'**: an **education department** in each province, **universities** (Calcutta, Bombay, Madras, **1857**), **primary schools in the vernacular, middle schools Anglo-vernacular, English colleges**, **grants-in-aid**, teacher training, **women's and vocational education**.
After 1857	**Hunter Commission (1882)**: stress on **primary and secondary** education, private enterprise; **Indian Universities Act 1904** (Curzon) and the **Raleigh Commission (1902)**: tighter government control. Result: a **small educated middle class** (lawyers, teachers, clerks) that later led **nationalism**, but **literacy stayed very low**, especially for **women**.
Press	**James Hickey's Bengal Gazette (1780)** was the first newspaper; **Ram Mohan Roy's Sambad Kaumudi (1821)**; the press became a **voice of nationalism** and was controlled by laws: **Licensing Regulations 1823 (Adam)**, **Metcalfe freed the press (1835)**, **Vernacular Press Act 1878**, **Newspaper (Incitement to Offences) Act 1908**.

4. Changes in society and culture

- **Law and rights:** a **uniform legal code** (**Indian Penal Code 1860**, Code of Criminal Procedure), **courts and the idea of equality before law** (though Europeans got special treatment); **social laws** against **sati (1829)**, for **widow remarriage (1856)**, against **female infanticide (1870)** and the **Age of Consent Act (1891)**.
- **New classes and ideas:** an **English-educated middle class**, a **new urban class** of clerks, lawyers and traders; ideas of **liberty, equality and nationalism** from the West; **missionaries** ran schools and hospitals; **socio-religious reform movements** (a separate topic) arose in reply.
- **Knowledge and institutions: Census** (first **1872**, regular from **1881**), **Archaeological Survey of India (1861**, Alexander Cunningham), **Geological Survey (1851)**, **Survey of India**, **medical colleges** (Calcutta and Madras, **1835**), **printing presses**; **Indo-Saracenic architecture** and 'Company painting'.
- **Colonial attitude:** British rulers kept a **racial distance** (clubs, railway compartments, the Ilbert Bill dispute), ruled with a **'divide and rule'** policy and described India as **backward**; Indians answered with **reform and, later, a national movement**.

5. The Andhra link

- **Famines:** the **Great Famine of 1876–78** (Telugu **'Dhatu' famine**, from the Telugu year **Dhatu**) hit the Madras Presidency's Telugu districts, from the Ceded districts and Nellore to the Circars, and Hyderabad

State; earlier the **Guntur famine of 1832–33** was also called a terrible disaster. They led to **famine relief works** and later the **Famine Code**.

- **Irrigation and crops: Cotton's anicuts** on the **Godavari (1852)** and **Krishna (1855)** raised paddy and trade in the delta, with **Cocanada (Kakinada)** growing as a port; **cotton, tobacco, indigo and groundnut** became **cash crops** in different regions, and **Masulipatnam**, once the great cloth port, **declined**.
- **Education and culture: Robert Noble** started a school at **Masulipatnam (1843)**, later **Noble College**; **Madras University (1857)** served the Telugu country; **C.P. Brown** (1798–1884, Madras Civil Service, worked in **Kadapa** and Madras) wrote the **Telugu–English dictionary (1852)**, collected and published **Vemana's verses** and old Telugu texts; **Colin Mackenzie** (first **Surveyor General of India**) gathered the **Mackenzie Manuscripts** and local histories (**kaifiyats**) from across the Telugu land with the help of **Kavali Venkata Borayya** and his brothers.

6. Impact

Area	Points
Politics	A **centralised state** controlled from London; **princes** reduced to **dependents**; **divide and rule**; **new legislative councils** gave Indians a first, small voice.
Society	**English-educated middle class**, **social laws**, **press**; but **low literacy** and **racial discrimination**.
Economy	**Drain of wealth**, **de-industrialisation**, **commercial crops**, **famines**; **railways and post** unified markets but served **British trade**.
Religion	Policy of **religious non-interference** (Queen's Proclamation) but **missionary work** and **social laws** provoked **reform and revival movements**.

EXAM TRAP

1858 Crown rule, Viceroy Canning, Secretary of State + Council of 15; **1 Nov 1858** Queen's Proclamation; **1861** Councils Act; **1870** Mayo; **1878** Vernacular Press Act, Arms Act (Lytton); **1882** Ripon: local self-government, Hunter Commission; **1883** Ilbert Bill; **1835** Macaulay's Minute; **1854** Wood's Despatch; **1857** three universities; **1853** first railway; **1859–60** Indigo Revolt; **1876–78** Great Famine; **1880** Strachey Famine Commission; **Naoroji 1867, 1901**, **Dutt 1901–03**; **Cotton: Godavari 1852, Krishna 1855**; **C.P. Brown 1852**; **Mackenzie manuscripts**.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements about the Government of India Act, 1858:
1. It transferred the government of India from the Company to the British Crown.
2. A Secretary of State for India assisted by a Council of 15 members was created.
3. The Board of Control and the Court of Directors were abolished.
4. It created a Legislative Council in India for the first time.
Which of the above are correct?

A. 1 and 2 only

B. 2 and 4 only

C. 1, 3 and 4 only

D. 1, 2 and 3 only

Answer: D. Legislative councils existed earlier (1833 and 1853); the 1858 Act changed the governing authority, not the council.

PRACTICE

Match List-I (work or event) with List-II (person):
List-I: A. Poverty and Un-British Rule in India B. Economic History of India C. Famine Commission of 1880 D. Vernacular Press Act 1878
List-II: 1. Richard Strachey 2. Lord Lytton 3. R.C. Dutt 4. Dadabhai Naoroji

A. A-3, B-4, C-1, D-2

B. A-4, B-3, C-1, D-2

C. A-4, B-3, C-2, D-1

D. A-4, B-1, C-3, D-2

Answer: B. Naoroji wrote Poverty and Un-British Rule; Dutt the Economic History; Strachey headed the 1880 Famine Commission; Lytton passed the 1878 Press Act.

PRACTICE

Which one of the following statements is NOT correct?

A. Macaulay's Minute of 1835 supported English as the medium of higher education.

B. Universities at Calcutta, Bombay and Madras were set up in 1857.

C. Wood's Despatch of 1854 recommended removing the vernacular languages from schools.

D. The Hunter Commission of 1882 emphasised primary and secondary education.

Answer: C. Wood's Despatch recommended vernacular primary schools, Anglo-vernacular middle schools and English colleges.

PRACTICE

Assertion (A): Dadabhai Naoroji's Drain of Wealth theory held that a part of India's wealth was transferred to Britain without return.
Reason (R): The Indian Civil Service examination was held only in London until 1922.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: B. Both are true, but the London-only examination does not explain the drain theory, which concerned Home Charges, profits and pensions.

PRACTICE

During the term of a Viceroy the Vernacular Press Act was repealed, the first Factory Act was passed, the Hunter Education Commission was appointed and a resolution on local self-government was issued. Who was this Viceroy?

A. Lord Lytton

B. Lord Curzon

C. Lord Dalhousie

D. Lord Ripon

Answer: D. These were measures of Lord Ripon (1880–84), the father of local self-government.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the 1858 Act, the Proclamation and the 1861 and 1892 Councils Acts. 2. Write the economy: drain, de-industrialisation, commercial crops, famines, railways, industry. 3. Write education (1835, 1854, 1857, 1882), press laws and the Andhra link.

POLITY · Legislative, administrative and financial relations between the Union and States

UNDERSTAND IT FIRST

What is it? Day 14 covered who makes laws (**legislative relations**, Arts 245–255) and the basic **executive** rule. Here we complete the picture: **administrative relations** (how the Centre and States **work together or control each other**) and **financial relations** (**who collects which taxes and who gets the money**). Both exist because **a State cannot run on its own tax income** and many problems (rivers, crime, trade) **cross State borders**. The Constitution keeps the **Centre financially strong** and then **shares money** through the **Finance Commission**.

1. Administrative relations (Part XI, Arts 256–263)

Provision	Details
Arts 256, 257	States must **comply with Union laws**; the Centre can **give directions** on **national and military communications**, **railways** and such matters; failure can lead to **Art 356** (via **Art 365**).
Art 258, 258A	**Delegation of functions** by consent: Centre ⇄ States.
Art 261	**Full faith and credit** to the public acts, records and judicial proceedings of the Union and every State.
Art 262 (water disputes)	**Parliament** can make law to **adjudicate disputes over inter-State rivers**, and can **bar the Supreme Court and other courts** from hearing them. **Inter-State River Water Disputes Act, 1956** sets up **tribunals**: e.g. **Krishna** (1969, **Bachawat**; **second 2004, Brijesh Kumar**), **Godavari** (1969), **Cauvery** (1990).
Art 263 (Inter-State Council)	The **President** can set up a council to **investigate and discuss** common interests; the **Inter-State Council** was set up in **1990** (on the **Sarkaria Commission's** advice) with the **Prime Minister as chair**, Chief Ministers as members.
Zonal Councils	Set up by the **States Reorganisation Act, 1956** (statutory, not constitutional); **five** (Northern, Central, Eastern, Western, **Southern**: Andhra Pradesh, Telangana, Karnataka, Kerala, Tamil Nadu, Puducherry); chaired by the **Union Home Minister**; advisory. The **North-Eastern Council (1971)** is separate.
All-India Services (Art 312)	**IAS, IPS and Indian Forest Service (1966)**: recruited by the **Union**, serve under **States**, controlled by the Centre; a new one needs a **Rajya Sabha resolution by two-thirds** (**All-India Services Act, 1951**).
Trade (Arts 301–307)	**Freedom of trade, commerce and intercourse** throughout India (Art 301), subject to reasonable restrictions by Parliament or States in the public interest; an **Inter-State Trade and Commerce Commission** (Art 307) is allowed but never set up.
Other links	**Art 355**: Union must protect every State from **external aggression and internal disturbance**; **Governor** appointed by the President; **single integrated judiciary**, **Election Commission** and **CAG**; the extra-constitutional **National Development Council (1952)** and, since **2015**, **NITI Aayog** (replaced the Planning Commission).

2. Financial relations (Part XII, Arts 264–293)

Union–State finance

Provision	Details
Taxing powers (Seventh Schedule)	**Union List (entries 82–92C): income tax** (not on agricultural income), **customs**, **corporation tax**, excise on tobacco and some goods. **State List (entries 45–63): land revenue**, **tax on agricultural income**, **stamp duty** (on documents not in the Union List), **excise on alcoholic liquor**, **taxes on land and buildings**, **electricity**, **entertainment**, **profession tax** (cap **₹2,500** a year, Art 276). **Residuary taxes** (e.g. gift tax) go to the **Union** (Art 248).
Art 268, 269, 269A	**268**: duties **levied by the Union but collected and kept by the States** (e.g. **stamp duties** on bills of exchange). **269**: taxes **levied and collected by the Union, assigned to States** (e.g. tax on **consignment of goods**). **269A** (after **GST**): **IGST on inter-State supply** is levied and collected by the **Union** and **shared** as the **GST Council** recommends.
Art 270 and **246A**	Taxes **levied and collected by the Union and shared** with States: **income tax** (except on agricultural income), and since the **101st Amendment (2016) CGST and IGST** (the Union's share of **GST**). States levy **SGST**. **Art 271**: Union **surcharge** on these taxes is **kept by the Union alone**.
GST Council (Art 279A)	Chair: **Union Finance Minister**; members: **Union Minister of State (Finance)** and the **State finance ministers**. Decisions need **three-fourths of weighted votes**: the **Centre has one-third** and **all States together two-thirds**; so neither can decide alone.
Finance Commission (Art 280)	Appointed by the **President every 5 years** (chair + 4 members). It recommends: (1) **sharing of net tax proceeds** between the Union and States, and among States; (2) the **principles of grants-in-aid** (Art 275); (3) measures to **supplement State funds for panchayats and municipalities**; (4) any other matter referred. Its **recommendations are advisory**, but the Government generally accepts them and places them before Parliament. **First FC 1951 (K.C. Neogy)**; **14th FC (Y.V. Reddy)**: **42%** of the divisible pool to States; **15th FC (N.K. Singh, 2021–26)**: **41%**; the **16th FC (chair: Arvind Panagariya)** is for **2026–31** (check its latest recommendations).
Grants (Art 275, 282)	**Art 275**: **statutory grants** from Parliament to needy States on the Finance Commission's advice (also for **Scheduled Tribe welfare**). **Art 282**: **discretionary grants** by the Union or a State for any **public purpose** (used for **centrally sponsored schemes**). **Special category status** was a **Planning Commission idea** (not in the Constitution) for hill and backward States.
Borrowing (Art 292, 293)	The **Union** can borrow **inside and outside India** on the security of the Consolidated Fund of India within limits set by Parliament; **States** can borrow **only within India**, and need the **Centre's consent** if a **loan to the State is still owed to the Centre**.
Other	**Art 289**: **State property and income are exempt from Union tax**, but the Union can tax a State's **trade or business**; **Art 285**: Union property is exempt from State tax; **Art 360**: **Financial Emergency**; **Art 300A**: **right to property** (a legal right).

- **Why 'vertical imbalance'?** The Centre collects most of the **major taxes** (income tax, corporation tax, customs), while **States have large spending duties** (health, police, agriculture, education). This gap is filled by **tax sharing and grants**, decided by the **Finance Commission**. **Horizontal imbalance** (richer and poorer States) is balanced by the **formulae** used among States (income distance, population, area, forest cover, demographic performance).
- **Sarkaria Commission (1983–88)** and **Punchhi Commission (2007–10)** studied these relations and urged **cooperative federalism**; **Art 274**: a State Bill affecting taxes in which States are interested needs the **President's prior recommendation**.

EXAM TRAP

Admin: Art 256–257 compliance, 258 delegation, 261 full faith, **262 river disputes (1956 Act; Krishna, Godavari, Cauvery tribunals)**, **263 Inter-State Council (1990)**, **Zonal Councils (1956)**, **Art 312 All-India Services**, 301–307 trade, 355. **Finance: 268** (levied by Union, kept by States), **269A** IGST, **270** shared, **271** surcharge Union only, **275** statutory grants, **280** Finance Commission (every 5 years, advisory), **282** discretionary grants, **292–293** borrowing, **279A** GST Council (3/4 weighted; Centre 1/3, States 2/3), **289**, **276** profession tax ₹2,500; **14th FC 42%, 15th FC 41%**.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements about the Finance Commission of India:
1. It is appointed by the President under Article 280, normally every five years.
2. It recommends how the net proceeds of taxes are shared between the Union and the States.
3. Its recommendations are binding on the Government of India.
4. The 15th Finance Commission was chaired by N.K. Singh.
Which of the above are correct?

A. 1 and 2 only

B. 2, 3 and 4 only

C. 1, 2 and 4 only

D. 1, 2, 3 and 4

Answer: C. The Finance Commission's recommendations are advisory, not binding.

PRACTICE

Match List-I (Article) with List-II (provision):
List-I: A. Article 268 B. Article 269A C. Article 270 D. Article 275
List-II: 1. Grants-in-aid to needy States 2. Duties levied by the Union but collected and kept by the

States 3. IGST on inter-State supply levied by the Union and apportioned 4. Taxes levied and collected by the Union and distributed between the Union and States

A. A-2, B-3, C-4, D-1

B. A-3, B-2, C-4, D-1

C. A-2, B-3, C-1, D-4

D. A-2, B-4, C-3, D-1

Answer: A. 268: collected by States; 269A: IGST; 270: shared taxes; 275: grants.

PRACTICE

Which one of the following statements is NOT correct?

A. Zonal Councils were set up by the States Reorganisation Act, 1956 and are chaired by the Union Home Minister.

B. Parliament can create a new All-India Service if the Rajya Sabha passes a resolution by a two-thirds majority.

C. A State can freely borrow from outside India without any restriction.

D. The Inter-State Council was set up in 1990 under Article 263.

Answer: C. States can borrow only within India (Art 293); only the Union can borrow abroad.

PRACTICE

Assertion (A): Decisions of the GST Council require a three-fourths majority of the weighted votes of members present and voting.
Reason (R): The Centre has one-third of the weighted votes and all the States together have two-thirds.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: A. Because of the 1/3 and 2/3 weights, no side can decide alone: R explains the three-fourths rule.

PRACTICE

Two riparian States quarrel over the sharing of the water of an inter-State river, and Parliament passes a law to adjudicate the dispute and bars the courts from hearing it. Which Article authorises this?

A. Article 263

B. Article 262

C. Article 280

D. Article 275

Answer: B. Art 262 lets Parliament adjudicate inter-State river disputes and bar court jurisdiction (the 1956 Act).

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the administrative provisions (Arts 256–263, 312, 355) with one line each. 2. Write the taxes in the Union and State Lists, Arts 268, 269A, 270, 271. 3. Write Art 280, 275, 282, 279A, borrowing, and the 14th and 15th Finance Commission shares.

MATHS · Time and distance

UNDERSTAND IT FIRST

What is it? These problems link **three quantities**: **distance (D)**, **speed (S)** and **time (T)**. The one rule behind everything: **Speed = Distance ÷ Time**, so **D = S × T** and **T = D ÷ S**. **At a fixed distance, more speed means less time** (they are **inversely** related). You must keep **units consistent**: **km with km/h and hours**, or **metres with m/s and seconds**. To convert **km/h to m/s multiply by 5/18**; for **m/s to km/h multiply by 18/5**.

Time and distance: toolkit

1. Average speed and relative speed

- **Average speed = total distance ÷ total time** (never the plain average of speeds). If a **same distance** is covered at **x** and then **y**, **average speed = 2xy/(x + y)**. **Example:** to and fro at **60** and **40 km/h**: 2 × 60 × 40 / 100 = **48 km/h** (not 50).
- **Relative speed:** two bodies **moving the same way**: **x − y**; **moving towards each other (opposite)**: **x + y**. **Example:** two cars 150 km apart approach at **50** and **25 km/h**: they meet in 150/75 = **2 hours**.

- **Late or early:** if a person at **5/6 of the usual speed** is late, the **time is 6/5 of the usual**: the **extra 1/5 of the usual time equals the delay**. **Example:** delay **15 min** ⇒ usual time = 5 × 15 = **75 min**.

2. Trains

- The train has **length**, so it must **travel its own length** more than a point object. **Crossing a pole or a person:** distance = **length of train**. **Crossing a platform or bridge:** distance = **train length + platform length**. **Example:** a **180 m** train at **54 km/h** (= 54 × 5/18 = **15 m/s**) crosses a **270 m** platform in (180 + 270)/15 = **30 s**.
- **Two trains:** distance to cross = **sum of both lengths**; speed = **relative speed** (**sum** if opposite, **difference** if the same way). **Example:** 120 m and 180 m trains at 54 and 36 km/h opposite: relative = 90 km/h = **25 m/s**; time = 300/25 = **12 s**.

3. Boats and streams

- A boat has its own speed in still water (**b**), and the **stream** has a speed (**s**). **Downstream** (with the current): **b + s**. **Upstream** (against it): **b − s**. If **downstream speed = d** and **upstream speed = u**: **b = (d + u)/2** and **s = (d − u)/2**.
- **Example:** a boat goes **20 km downstream in 2 hours** (d = 10 km/h) and returns **upstream in 4 hours** (u = 5 km/h): **b = 7.5 km/h**, **s = 2.5 km/h**.

EXAM TRAP

S = D/T; km/h × 5/18 = m/s; average speed (equal distances) = 2xy/(x + y); relative speed: same way x − y, opposite x + y; train and pole: its length; train and platform: both lengths; two trains: sum of lengths; boat: down b + s, up b − s, b = (d + u)/2, s = (d − u)/2; at 5/6 speed time is 6/5.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

A man travels from A to B at 60 km/h and returns from B to A along the same route at 40 km/h. What is his average speed for the whole journey?

A. 48 km/h

B. 50 km/h

C. 45 km/h

D. 52 km/h

Answer: A. 2xy/(x + y) = 2 × 60 × 40/100 = 48 km/h.

PRACTICE

A train 180 m long running at 54 km/h crosses a platform 270 m long. How much time does it take?

A. 20 seconds

B. 25 seconds

C. 30 seconds

D. 36 seconds

Answer: C. Speed = 15 m/s; distance = 180 + 270 = 450 m; time = 30 s.

PRACTICE

Two trains of lengths 120 m and 180 m run in opposite directions at 54 km/h and 36 km/h. In how many seconds will they completely cross each other?

A. 10 seconds

B. 15 seconds

C. 20 seconds

D. 12 seconds

Answer: D. Relative speed = 90 km/h = 25 m/s; distance = 300 m; time = 12 s.

PRACTICE

A boat goes 20 km downstream in 2 hours and returns the same distance upstream in 4 hours. What are the speed of the boat in still water and the speed of the stream?

A. 8 km/h and 2 km/h

B. 7.5 km/h and 2.5 km/h

C. 7 km/h and 3 km/h

D. 6 km/h and 4 km/h

Answer: B. d = 10, u = 5; b = (10 + 5)/2 = 7.5; s = (10 − 5)/2 = 2.5 km/h.

PRACTICE

A person walking at 5/6 of his usual speed reaches his office 15 minutes late. What is his usual time to reach the office?

A. 60 minutes

B. 90 minutes

C. 75 minutes

D. 45 minutes

Answer: C. New time = 6/5 of usual; extra 1/5 of usual time = 15 min, so usual time = 75 min.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write S = D/T, unit conversion and the average speed formula with one example. 2. Write relative speed and the three train cases. 3. Write the boat formulas and solve the 'late by 15 minutes' example.

NOTEBOOK PAGE · Day 15: write this from memory

Close the notes. Write these headings, fill every blank, then correct in a different colour.

HISTORY: Changes under the British

Area	Write
Administration	1858 Act, Viceroy, Secretary of State, Queen's Proclamation, 1861 Councils Act, Mayo 1870, Ripon 1882
Economy	Drain (Naoroji, Dutt), de-industrialisation, commercial crops, railways, famines (1876–78, Strachey 1880), modern industry
Education and press	Macaulay 1835, Wood 1854, universities 1857, Hunter 1882; Hickey 1780, Vernacular Press Act 1878
Society and culture	Social laws, census, ASI, medical colleges, new middle class
Andhra link	Famine of 1876–78, Cotton's irrigation, C.P. Brown, Mackenzie manuscripts, Noble's school 1843

POLITY: Administrative and financial relations

- Administrative: Arts 256–263, 355, 356, 312 (All-India Services), 262 (water disputes), 263 (Inter-State Council), Zonal Councils, NITI Aayog.
- Financial: Arts 268–270, 246A, 275, 280, 282, 292–293, 279A (GST Council); Finance Commission (14th 42%, 15th 41%); borrowing rules; Art 289.

MATHS: Time and distance

Write: S = D/T; km/h ↔ m/s; average speed 2xy/(x + y); relative speed (same, opposite); train formulas; boats (down, up, still water, stream); one worked example of each.